

Series2(Solution)

Exercise1:

1. Convert these binary numbers to decimal.

$$1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$$

$$= 16 + 0 + 4 + 2 + 0 = 22$$

- (a) $10110_2 = 22_{10}$
- (b) $01101011_2 = 107_{10}$
- (c) $1111010111_2 = 987_{10}$
- (d) $10010101_2 = 149_{10}$
- (e) $11111111_2 = 255_{10}$
- (f) $11011111_2 = 223_{10}$
- (g) $100100001001_2 = 2305_{10}$
- (h) $01101111_2 = 111_{10}$

2. Convert the following decimal values to binary.

(a) 37_{10}

1. Divide by 2:

$$37 \div 2 = 18 \text{ remainder } 1$$

$$18 \div 2 = 9 \text{ remainder } 0$$

$$9 \div 2 = 4 \text{ remainder } 1$$

$$4 \div 2 = 2 \text{ remainder } 0$$

$$2 \div 2 = 1 \text{ remainder } 0$$

$$1 \div 2 = 0 \text{ remainder } 1$$

2. Binary: Read the remainders from bottom to top:

Binary Value: 100101_2

- $37_{10} = 100101_2$
- $1000_{10} = 111101000_2$
- $205_{10} = 11001101_2$
- $13_{10} = 1101_2$
- $77_{10} = 1001101_2$
- $2133_{10} = 100001110101_2$
- $189_{10} = 10111101_2$
- $390_{10} = 110000110_2$
- $511_{10} = 11111111_2$
- $3410_{10} = 110101101010_2$

3. Convert the binary numbers in Problem 5-1 to octal and hex.

Binary to Octal Conversions

1. $10110_2 = 26_8$
2. $01101011_2 = 133_8$
3. $1111010111_2 = 375_8$
4. $10010101_2 = 125_8$
5. $11111111_2 = 377_8$
6. $11011111_2 = 337_8$
7. $100100001001_2 = 2211_8$
8. $01101111_2 = 67_8$

Binary to Hexadecimal Conversions

1. $10110_2 = 1A_{16}$
2. $01101011_2 = 6B_{16}$
3. $1111010111_2 = F7_{16}$
4. $10010101_2 = 95_{16}$
5. $11111111_2 = FF_{16}$
6. $11011111_2 = DF_{16}$
7. $100100001001_2 = 221_{16}$
8. $01101111_2 = 6F_{16}$

Exercise2: Convert the following binary numbers to their equivalent decimal values.

(a) $1100_2 = 12_{10}$

(b) $1001.1001_2 = 9.5625_{10}$

(c) $10011011001.10110_2 = 1241.6875_{10}$

(d) $10011_2 = 19_{10}$

(e) $1100.0101_2 = 12.3125_{10}$

(f) $10011100100.10010_2 = 1218.5625_{10}$

Exercise3:

1- Convert each hex number to its decimal equivalent.

(a) 743 (16)

$$7 \times 16^2 + 4 \times 16^1 + 3 \times 16^0 = 7 \times 256 + 4 \times 16 + 3 \times 1 = 1792 + 64 + 3 = 1859$$

- Decimal: 1859

(b) 2000 (16)

$$2 \times 16^3 + 0 \times 16^2 + 0 \times 16^1 + 0 \times 16^0 = 2 \times 4096 + 0 + 0 + 0 = 8192$$

- Decimal: 8192

(a) $743 (16) = 1859 (10)$

(b) $2000 (16) = 8192 (10)$

(c) $7FF (16) = 2047 (10)$

(d) $36 (16) = 54 (10)$

(e) $55 (16) = 85 (10)$

(f) $165 (16) = 357 (10)$

(g) $1204 (16) = 4612 (10)$

(h) $37FD (16) = 14333 (10)$

(i) $ABCD (16) = 43881 (10)$

(j) $000F (16) = 15 (10)$

2- Convert each of the following decimal numbers to hex.

Example 2: Convert 1024 (decimal) to Hexadecimal

1. Divide the decimal number by 16:

- $1024 \div 16 = 64$ (quotient)
- Remainder: $1024 - (16 \times 64) = 1024 - 1024 = 0$

2. Record the remainder:

- The remainder is 0. This is the least significant digit (rightmost) in hexadecimal.

3. Continue dividing the quotient by 16:

- $64 \div 16 = 4$ (quotient)
- Remainder: $64 - (16 \times 4) = 64 - 64 = 0$

4. Record the next remainder:

- The remainder is 0.

5. Continue dividing the quotient by 16:

- $4 \div 16 = 0$ (quotient)
- Remainder: $4 - (16 \times 0) = 4$

6. Record the final remainder:

- The remainder is 4. This is the most significant digit (leftmost).

Putting it All Together

- The remainders, read from last to first, give you the hexadecimal representation.
- So, $1024_{10} = 400_{16}$.

Example 3: Convert 65,536 (decimal) to Hexadecimal

1. Divide the decimal number by 16:

- $65,536 \div 16 = 4096$ (quotient)
- Remainder: $65,536 - (16 \times 4096) = 65,536 - 65,536 = 0$

2. Record the remainder:

- The remainder is 0. This is the least significant digit (rightmost) in hexadecimal.

3. Continue dividing the quotient by 16:

- $4096 \div 16 = 256$ (quotient)
- Remainder: $4096 - (16 \times 256) = 4096 - 4096 = 0$

4. Record the next remainder:

- The remainder is 0.

5. Continue dividing the quotient by 16:

- $256 \div 16 = 16$ (quotient)
- Remainder: $256 - (16 \times 16) = 256 - 256 = 0$

6. Record the next remainder:

- The remainder is 0.

7. Continue dividing the quotient by 16:

- $16 \div 16 = 1$ (quotient)
- Remainder: $16 - (16 \times 1) = 16 - 16 = 0$

8. Record the final remainder:

- The remainder is 0.

9. Continue dividing the quotient by 16:

- $1 \div 16 = 0$ (quotient)
- Remainder: $1 - (16 \times 0) = 1$

10. Record the last remainder:

- The remainder is 1. This is the most significant digit (leftmost).

Putting it All Together

- The remainders, read from last to first, give you the hexadecimal representation.
- So, $65,536_{10} = 10000_{16}$.

$$59 (10) = 3B (16)$$

$$1024 (10) = 400 (16)$$

$$65,536 (10) = 10000 (16)$$

$$372 (10) = 174 (16)$$

$$771 (10) = 303 (16)$$

$$255 (10) = FF (16)$$

$$919 (10) = 398 (16)$$

$$2313 (10) = 90B (16)$$

$$25,619 (10) = 63B3 (16)$$

$$2048 (10) = 800 (16)$$

3-Take each four-bit binary number in the order they are written and write the equivalent hex digit

- (a) 1001 (2) = 9 (16)
- (b) 1101 (2) = D (16)
- (c) 1111 (2) = F (16)
- (d) 0010 (2) = 2 (16)
- (e) 1100 (2) = C (16)
- (f) 1011 (2) = B (16)
- (g) 0001 (2) = 1 (16)
- (h) 0101 (2) = 5 (16)

4-Convert the hex values in Problem 3-1 to binary

(a) 743 (16)

- 7 = 0111
- 4 = 0100
- 3 = 0011

Binary: 0111 0100 0011

(b) 2000 (16)

- 2 = 0010
- 0 = 0000
- 0 = 0000
- 0 = 0000

Binary: 0010 0000 0000 0000

- (a) 743 (16) = 0111 0100 0011 (2)
- (b) 2000 (16) = 0010 0000 0000 0000 (2)
- (c) 7FF (16) = 0111 1111 1111 (2)
- (d) 36 (16) = 0011 0110 (2)
- (e) 55 (16) = 0101 0101 (2)
- (f) 165 (16) = 0001 0110 0101 (2)
- (g) 1204 (16) = 0001 0010 0000 0100 (2)
- (h) 37FD (16) = 0011 0111 1111 1101 (2)
- (i) ABCD (16) = 1010 1011 1100 1101 (2)
- (j) 000F (16) = 0000 0000 0000 1111 (2)

5- List the hex numbers in sequence from 280 to 2A0.

The hex numbers from 280 (16) to 2A0 (16) are:

- | | |
|-------|-------|
| • 280 | • 28F |
| • 281 | • 290 |
| • 282 | • 291 |
| • 283 | • 292 |
| • 284 | • 293 |
| • 285 | • 294 |
| • 286 | • 295 |
| • 287 | • 296 |
| • 288 | • 297 |
| • 289 | • 298 |
| • 28A | • 299 |
| • 28B | • 29A |
| • 28C | • 29B |
| • 28D | • 29C |
| • 28E | • 29D |
| | • 29E |
| | • 29F |
| | • 2A0 |

6- How many hex digits are required to represent decimal numbers up to 1 million?

To find how many hex digits are needed to represent decimal numbers up to 1 million (1,000,000), we can use the following formula:

1. Convert 1,000,000 to hexadecimal:

$$1,000,000 \div 16 = 62,500 \quad (\text{remainder: } 0)$$

$$62,500 \div 16 = 3,906 \quad (\text{remainder: } 4)$$

$$3,906 \div 16 = 244 \quad (\text{remainder: } 2)$$

$$244 \div 16 = 15 \quad (\text{remainder: } 4)$$

$$15 \div 16 = 0 \quad (\text{remainder: } 15, \text{ which is F in hex})$$

So, $1,000,000_{10} = F4240_{16}$.

2. The number of hex digits in F4240 is 5.

Thus, 5 hex digits are required to represent decimal numbers up to 1 million.

Exercise4:

Part A: Convert the following binary numbers to their hexadecimal and octal equivalents

(a) 10100100

- Hexadecimal:

- Group into 4 bits: 1010 0100
- Convert to hex: A4

- Octal:

- Group into 3 bits: 101 001 00
- Convert to octal: 244

(b) 10100111

- Hexadecimal:

- Group into 4 bits: 1010 0111
- Convert to hex: A7

- Octal:

- Group into 3 bits: 101 001 111
- Convert to octal: 247

(c) 10100,010

- Hexadecimal:

- Group into 4 bits: 0001 0100 and 0000 0010
- Convert to hex: 14.2

- Octal:

- Group into 3 bits: 001 010 000 and 000 010
- Convert to octal: 24.2

(d) 10111,0101

- Hexadecimal:

- Group into 4 bits: 1011 1010 and 0101
- Convert to hex: BA.5

- Octal:

- Group into 3 bits: 101 111 010 and 0101
- Convert to octal: 572.5

(e) 11011111000

- Hexadecimal:

- Group into 4 bits: 0001 1011 1110 00
- Convert to hex: 1BE0

- Octal:

- Group into 3 bits: 011 011 111 000
- Convert to octal: 3370

(f) 11000,01011

- Hexadecimal:

- Group into 4 bits: 1100 0010 and 1011
- Convert to hex: C2.B

- Octal:

- Group into 3 bits: 011 000 001 and 011
- Convert to octal: 301.3

(g) 110011100111

- Hexadecimal:
 - Group into 4 bits: 1100 1110 0111
 - Convert to hex: CE7
- Octal:
 - Group into 3 bits: 110 011 100 111
 - Convert to octal: 6347

(h) 100110110,101011

- Hexadecimal:
 - Group into 4 bits: 1001 1011 and 0101 011
 - Convert to hex: 9B.2B
- Octal:
 - Group into 3 bits: 100 110 110 and 101 011
 - Convert to octal: 466.53

Part B: Convert the fractional decimal numbers to binary, octal, and hexadecimal forms

(a) $(0.625)_{10}$

- Binary: Multiply 0.625 by 2:
 - $0.625 \times 2 = 1.25 \rightarrow 0.1$
 - $0.25 \times 2 = 0.5 \rightarrow 0.01$
 - $0.5 \times 2 = 1.0 \rightarrow 0.001$
 - Binary: 0.101
- Octal: Convert 0.101 to octal by grouping into 3 bits:
 - Octal: 0.5
- Hexadecimal: Convert 0.101 to hex by grouping into 4 bits:
 - Hex: 0.A

(b) $(0.450)_{10}$

- Binary: Multiply 0.450 by 2:
 - $0.450 \times 2 = 0.9 \rightarrow 0.0$
 - $0.9 \times 2 = 1.8 \rightarrow 0.1$
 - Continue similar steps...
 - Binary: 0.01110011 (approximately)
- Octal: Group the binary in 3s:
 - Octal: 0.346 (approximately)
- Hexadecimal: Group the binary in 4s:
 - Hex: 0.73 (approximately)

(f) $(0.350)_{10}$

- Binary: Multiply 0.350 by 2 and continue:
 - Binary: 0.010110
- Octal: Convert to octal:
 - Octal: 0.264
- Hexadecimal: Convert to hex:
 - Hex: 0.5A

(c) $(2.500)_{10}$

- Binary: Convert 2 to binary: 10. Then convert 0.500:
 - $0.500 \times 2 = 1.0$
 - Binary: 10.1
- Octal: Convert 10.1 to octal by grouping:
 - Octal: 2.4
- Hexadecimal: Convert 10.1 to hex:
 - Hex: 2.8

(d) $(5.1625)_{10}$

- Binary: Convert 5 to binary: 101. Then convert 0.1625:
 - Multiply 0.1625 by 2 and continue...
 - Binary: 101.00101010 (approximately)
- Octal: Convert to octal:
 - Octal: 5.124
- Hexadecimal: Convert to hex:
 - Hex: 5.29

(e) $(7.1875)_{10}$

- Binary: Convert 7 to binary: 111. Then convert 0.1875:
 - Multiply 0.1875 by 2 and continue...
 - Binary: 111.0011
- Octal: Convert to octal:
 - Octal: 7.14
- Hexadecimal: Convert to hex:
 - Hex: 7.3

Exercise5: Perform the conversions to the appropriate units for the following values:

1. $(87)_{10}$
- Binary: 1010111_2
 - Hexadecimal: 57_{16}

2. $(328)_{10}$
- Binary: 101001000_2
 - Hexadecimal: 148_{16}

3. $(1900)_{10}$
- Binary: 11101100100_2
 - Hexadecimal: $76C_{16}$

4. $(3E)_{16}$
- Binary: 00111110_2
 - Decimal: 62_{10}

5. $(234)_{10}$
- Binary: 001000110100_2
 - Decimal: 564_{10}

6. $(AF3)_{16}$
- Binary: 101011110011_2
 - Decimal: 2819_{10}

7. $(10011100)_2$

- Decimal: 156_{10}
- Hexadecimal: $9C_{16}$

8. $(11110101)_2$

- Decimal: 245_{10}
- Hexadecimal: $F5_{16}$

9. $(10010101101)_2$

- Decimal: 1213_{10}
- Hexadecimal: $4B5_{16}$

10. $(34)_8$

- Binary: 011100_2
- Decimal: 28_{10}

11. $(567)_8$

- Binary: 101110111_2
- Decimal: 375_{10}

12. $(1534)_8$

- Binary: 001010101011100_2
- Decimal: 834_{10}

Exercise6: Perform the following operations in binary

$$1. (1101010)_2 + (1110011)_2 = (11011101)_2$$

$$2. (1010)_2 + (111)_2 = (10001)_2$$

$$3. (1010.011)_2 + (111.11)_2 = (10010.001)_2$$

$$4. (111011)_2 - (11)_2 = (111000)_2$$

$$5. (100001)_2 - (11110)_2 = (11)_2$$

$$6. (1001.1110)_2 - (11.1001)_2 = (110.011)_2$$

Exercise7: Perform the following operations in the binary system, knowing that the numbers are represented on a byte (8 bits). In the case of division, specify the remainder.

a) $10101+1010=1\ 1111$

b) $10111+1010=10\ 0001$

c) $110111+100111=101\ 1110$

d) $10101-1010=1011$

e) $10111-1010=1101$

f) $110111-100111=1\ 0000$

g) $100011*110=1101\ 0010$

h) $110110*101=1\ 0000\ 1110 = (14)_{10} \rightarrow \text{dépassement}$

i) $100101/101=111\ (\text{le reste} = 10)$

j) $11010111/110=10\ 0011\ (\text{le reste} = 101)$